PROJECT PROPOSAL

<The 15th e-ICON World Contest>

Mobile APP Name (Title)	TrashIQ		
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□ Summary of Mobile App

TrashIQ is designed to support **SDG 13 (Climate Action)** by promoting smart waste management and environmental awareness. Users can register, create profiles, and begin tracking their contributions to sustainability. The **Home Page** provides educational content on climate change, recycling, and sustainable living. A **Report Page** features a circular tracker which displays recyclable non-recyclable, and biomedical materials the user has reported using our system.

Users earn **eco-points** for properly sorting and submitting waste using our system, which motivates continued engagement because many environmental protection bodies in a country provide certificates and some prizes to those who help in environmental protection so these bodies will use our application to reward a user with more points which will motivate them. Also an integrated **AI scanner tool** allows users to take photos of waste items to identify whether they are recyclable or not which makes easy to separate and easy when collecting. The app includes a **map feature** that highlights waste zones—**green** for well-managed areas and **red** for areas with high non-recyclable waste or high accumulation of waste. It also helps users locate nearby recycling and waste collection centers.

The **Profile Page** shows user information, accumulated points, and personal impact stats. Overall, the app combines education, technology, and gamification to make climate action accessible and rewarding for everyone.

□ Phenomenon or problems related to the theme of the contest that you want to realize with SW/Mobile App

The app addresses the critical issue of **improper waste management**, a major contributor to climate change, in line with **SDG 13 (Climate Action)**. It supports both individuals and waste collectors by identifying and categorizing non-recyclable materials, helping collectors efficiently gather and transport them to proper facilities for reuse or safe disposal. This reduces environmental waste accumulation and promotes sustainable waste practices that directly contribute to climate action goals. It also helps in education awareness and location based tracking.

□ Design of algorithm(mechanism) and UI(User Interface) for SW/Mobile App

Algorithm

TrashIQ is based on image recognition (AI/ML model) technology to identify waste items captured by the user's camera and classify them as recyclable non-recyclable or medical wastes. Once classified, the user makes a report including names and location after making a report, updates are made on his/her profile including earning eco-points. The app also collects GPS location data to map waste trends in communities, highlighting zones with high recyclable or non-recyclable waste. Data from reports is also shared with registered waste collectors and environmental protection bodies in our country which may help them in planning and making reports.

User Interface

It consists of home page where the user finds about education awareness about climate change and waste management. It have report feature where the users sees his/her contribution on reporting recyclable and non-recyclable wastes in that section the user also scans waste and can make a report so that we can tell specialized bodies to go and collect those wastes and update the map
We have also map feature where the user can see areas with accumulated waste and waste collectors can know where to start meaning areas in risk of high waste accumulation we have also profile showing the user info and points earned

□ Development Tool/Program/Other necessary technology that would like to use

The app will be built using **React Native** for cross-platform compatibility. We will integrate **TensorFlow** for waste classification and **Google Maps API** will enable geolocation features, while **Figma** will help us in designing.

□ Impact of Mobile App on Teaching & Learning Activities

- Promotes **interactive and practical learning** by allowing users to scan, sort, and report waste in real-world making them to know recyclable or non-recyclable wastes.
- Supports **climate change education** through engaging content aligned with **SDG 13** example on the home page the user can learn about climate change.
- Encourages **behavioral change and environmental responsibility** through a gamified point and reward system.

※ **Project Proposal** should be submitted via e-ICON website (**PDF format only**)

<Notice>

1. Each team should submit **one Project Proposal** **per team.**
2. All mobile apps shall be developed in the **Android environment.**
3. The app development program will be suggested during the preliminaries and participants may use any program of their choice **except App Builder**.
4. The format of the project proposal may change for the final evaluation, in which case the changes will be communicated via the e-ICON website.
5. If necessary, you can use **additional pages for the proposal**.